

CAPE ST JAMES, BC, Canada

WMO: 711070

Lat: 51.9358N

Lon: 131.0158W

Elev: 293

StdP: 14.54

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.7	31.8	14.7	11.7	30.7	18.8	14.3	33.4	56.4	41.3	51.1	40.4	18.6	60	0.915

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	66.5	59.6	64.0	58.5	61.9	57.3	61.5	65.0	59.9	62.9	58.5	61.1	10.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	77.7	63.1	58.4	73.7	61.1	57.2	70.6	59.6	27.5	65.3	26.4	63.1	25.5	61.3	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
47.7	42.3	37.5	DB	25.5	72.6	6.4	3.7	20.9	75.2	17.1	77.3	13.5	79.4	8.8	82.0
			WB	21.4	64.8	5.7	2.4	17.2	66.6	13.9	68.0	10.7	69.4	6.5	71.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	42.1	41.9	42.2	45.3	49.0	52.8	56.2	57.9	56.0	50.5	45.3	42.3
	DBStd	6.79	4.73	3.82	3.69	3.14	3.22	3.19	2.51	2.51	2.75	2.99	3.74	4.22
	HDD50	1335	245	227	244	145	56	6	0	0	0	30	144	238
	HDD65	6024	709	647	708	590	496	366	272	219	271	451	592	703
	CDD50	788	0	0	1	5	25	91	194	246	180	44	2	0
	CDD65	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH74	5	0	0	0	0	0	2	3	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	18.6	21.8	20.1	19.8	18.5	16.3	16.4	15.8	15.4	15.6	19.6	21.4	22.2
Precipitation	PrecAvg	63.0	7.1	5.6	5.3	4.6	3.7	3.3	2.6	3.2	4.8	8.0	7.7	7.2
	PrecMax	82.4	12.0	11.0	9.0	9.3	7.9	6.5	7.1	6.7	11.0	12.4	12.7	12.0
	PrecMin	45.1	0.8	1.0	1.5	1.6	1.3	0.7	0.3	0.2	0.8	4.2	3.9	3.5
	PrecStd	8.9	2.2	2.0	1.9	1.8	1.7	1.5	1.5	1.8	2.2	2.0	2.3	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	51.2	53.0	59.1	63.1	70.4	69.1	70.0	67.5	60.9	53.5	50.6
		MCWB	49.5	46.5	47.0	47.3	53.6	58.3	60.2	63.4	59.0	54.5	51.1	48.8
	2%	DB	50.0	48.8	49.6	55.2	59.2	63.3	64.6	66.6	64.1	57.4	51.9	49.2
		MCWB	48.5	45.8	45.8	47.6	51.5	56.3	59.9	61.0	57.6	53.8	50.0	47.6
	5%	DB	48.5	47.5	48.0	52.0	56.2	59.6	62.1	64.1	61.6	55.7	50.9	48.2
		MCWB	46.9	45.1	45.2	46.6	50.7	55.1	58.2	59.6	56.5	53.0	49.3	46.8
	10%	DB	47.0	46.4	46.6	49.5	53.7	57.2	60.2	61.8	59.5	54.4	50.0	47.2
		MCWB	45.6	44.2	44.4	45.5	49.4	53.9	57.0	58.2	55.9	52.1	48.6	45.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	48.6	49.2	51.3	55.9	60.3	63.0	64.5	61.2	56.8	52.5	49.8
		MCDB	50.5	49.8	51.0	55.5	61.1	67.1	67.3	68.7	64.9	57.9	52.7	50.1
	2%	WB	48.2	46.8	47.7	49.3	53.4	57.5	60.4	62.3	59.1	55.2	50.9	48.3
		MCDB	48.7	48.0	49.1	52.3	57.9	61.9	63.7	66.0	62.1	56.3	51.2	48.8
	5%	WB	46.8	45.7	46.3	47.8	51.6	55.9	59.0	60.6	57.8	54.0	49.9	47.3
		MCDB	47.2	47.0	47.5	50.5	55.4	59.3	62.0	63.3	60.1	55.2	50.4	47.9
	10%	WB	45.9	44.8	45.0	46.3	50.3	54.5	57.6	59.1	56.8	52.7	49.0	46.2
		MCDB	46.4	46.0	46.3	49.0	53.2	57.3	60.0	61.5	58.8	54.0	49.8	47.0
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.5	6.3	7.0	8.2	8.0	8.1	7.9	7.2	6.0	5.2	4.8
		MCDBR	4.2	6.5	7.8	11.3	12.8	12.6	11.4	11.9	10.7	7.9	5.3	4.6
		MCWBR	4.1	5.5	6.2	8.2	9.3	9.0	8.3	8.4	7.7	6.1	5.0	5.0
	5% WB	MCDBR	4.3	5.8	6.4	9.8	11.8	11.7	11.3	11.3	9.0	6.6	4.6	4.4
		MCWBR	4.4	5.2	5.8	7.7	8.8	8.8	8.5	8.4	7.0	5.9	4.8	5.2
Clear-Sky Solar Irradiance	taub	0.319	0.325	0.351	0.388	0.385	0.369	0.368	0.370	0.369	0.349	0.327	0.310	
	taud	2.401	2.480	2.415	2.290	2.363	2.459	2.516	2.514	2.471	2.438	2.361	2.357	
	Ebn at Noon	218	255	268	269	275	281	279	273	259	239	209	198	
	Edh at Noon	19	23	30	37	36	34	31	30	28	24	19	17	
All-Sky Solar Radiation	RadAvg	208	483	826	1301	1625	1717	1648	1409	1021	571	279	175	
	RadStd	26	72	90	63	117	144	154	117	91	61	26	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page